Unit IV: Special Random Processes

Poisson, Pure Birth, Yule–Furry,
Birth and Death, and Renewal Processes

Dr. S. Balaji

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1 Introduction

In this section we discuss some special classes of random processes, in which we have initially the **Poisson process**.

The Poisson process is an example of a *discrete random process*, i.e., the state space S is discrete and the time set T is continuous.

We know that the pdf of the Poisson distribution for x number of successes is

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}, \qquad x = 0, 1, 2, \ldots$$

Here λ is a parameter.

1.1 Postulates of the Poisson process

Throughout, $o(h)$ denotes any function of h such that

$$\lim_{h\to 0} \frac{o(h)}{h} = 0,$$

i.e., $o(h)$ tends to zero faster than h (for example $h^2, h^3, \ldots$). Note that $o(h) \pm o(h) = o(h)$ and $c \cdot o(h) = o(h)$ for any constant c.

Definition. Let $X(t)$ represent the number of occurrences of a certain event in $(0, t)$. Then the discrete random process $\{X(t) : t \geq 0\}$ is a **Poisson process** with rate $\lambda > 0$ provided the following postulates are satisfied:

(i) $P\big(1 \text{ occurrence in } (t, t+h)\big) = \lambda h + o(h)$.

(ii) $P\big(2 \text{ or more occurrences in } (t, t+h)\big) = o(h)$.

(iii) $P\big(0 \text{ occurrence in } (t, t+h)\big) = 1 - \lambda h + o(h)$.

(iv) *(Independent increments)* The numbers of occurrences in non-overlapping time intervals are independent; in particular, the number of occurrences in $(t, t+h)$ is independent of $X(t)$, the number of occurrences in $(0, t)$.

(v) *(Stationary increments)* The probability that the event occurs a specified number of times in $(t_1, t_1 + t)$ depends only on the length t and not on t_1.

Also $X(0) = 0$.

Note. Postulate (iii) is not an extra assumption; it follows from (i) and (ii):

$$\begin{aligned} P(0 \text{ occ.}) &= 1 - P(1 \text{ occ.}) - P(2 \text{ or more occ.}) \\ &= 1 - \big(\lambda h + o(h)\big) - o(h) \\ &= 1 - \lambda h + o(h). \end{aligned}$$

1.2 Probability law of the Poisson process

In a Poisson process, the number of occurrences $X(t)$ in a time interval $(0, t)$ is given by the pdf (derived from the postulates in Section 9)

Poisson Process

$$P\big(X(t) = n\big) = \frac{e^{-\lambda t}(\lambda t)^n}{n!}, \qquad n = 0, 1, 2, \ldots$$

Here λt is the parameter of the Poisson process, and λ is the average rate of occurrences (per unit time).

Note. The above pdf will be useful to determine the number of customers arriving at a counter in a time interval of t minutes, the number of messages arriving at a telegraph office in t hours, etc.

2 Mean and Variance of the Poisson Process

2.1 Result 1: Mean of the Poisson process

$$\begin{aligned}
\text{Mean} = E\big(X(t)\big) &= \sum_n n\, P\big(X(t) = n\big) \\
&= \sum_{n=0}^{\infty} n\, \frac{e^{-\lambda t}(\lambda t)^n}{n!} \\
&= e^{-\lambda t} \sum_{n=1}^{\infty} \frac{n\,(\lambda t)^n}{n\,(n-1)!} \\
&= e^{-\lambda t} \sum_{n=1}^{\infty} \frac{(\lambda t)^n}{(n-1)!} \\
&= e^{-\lambda t} \left[\frac{(\lambda t)^1}{0!} + \frac{(\lambda t)^2}{1!} + \frac{(\lambda t)^3}{2!} + \cdots\right] \\
&= e^{-\lambda t}\, \lambda t \left[1 + \frac{\lambda t}{1!} + \frac{(\lambda t)^2}{2!} + \cdots\right] \\
&= e^{-\lambda t}\, \lambda t\, e^{\lambda t} \\
&= \lambda t.
\end{aligned}$$

$$\text{Mean of the Poisson process} = E\big(X(t)\big) = \lambda t.$$

Note. In the Poisson process, the mean value itself is a function of time t (namely λt). Hence the Poisson process is **not a stationary random process**.

2.2 Result 2: Variance of the Poisson process

We know that

$$\text{Var}\,\big(X(t)\big) = E\big(X^2(t)\big) - \big[E\big(X(t)\big)\big]^2.$$

Now,

$$\begin{aligned}
E\big(X^2(t)\big) &= \sum_n n^2\, P\big(X(t)=n\big) \\
&= \sum_{n=0}^{\infty} n^2\, \frac{e^{-\lambda t}(\lambda t)^n}{n!} \\
&= e^{-\lambda t} \sum_{n=0}^{\infty} \frac{n^2(\lambda t)^n}{n!} \\
&= e^{-\lambda t}\left[0 + \frac{1^2(\lambda t)^1}{1!} + \frac{2^2(\lambda t)^2}{2!} + \frac{3^2(\lambda t)^3}{3!} + \frac{4^2(\lambda t)^4}{4!} + \cdots\right] \\
&= e^{-\lambda t}\left[\lambda t + \frac{2(\lambda t)^2}{1!} + \frac{3(\lambda t)^3}{2!} + \frac{4(\lambda t)^4}{3!} + \cdots\right] \\
&= e^{-\lambda t}\,\lambda t\left[1 + \frac{2(\lambda t)}{1!} + \frac{3(\lambda t)^2}{2!} + \frac{4(\lambda t)^3}{3!} + \cdots\right] \\
&= e^{-\lambda t}\,\lambda t\left[1 + \frac{(1+1)(\lambda t)}{1!} + \frac{(2+1)(\lambda t)^2}{2!} + \frac{(3+1)(\lambda t)^3}{3!} + \cdots\right] \\
&= e^{-\lambda t}\,\lambda t\Bigg[\left(1 + \frac{\lambda t}{1!} + \frac{(\lambda t)^2}{2!} + \frac{(\lambda t)^3}{3!} + \cdots\right) \\
&\qquad + \left(\frac{\lambda t}{1!} + \frac{2(\lambda t)^2}{2!} + \frac{3(\lambda t)^3}{3!} + \cdots\right)\Bigg] \\
&= e^{-\lambda t}\,\lambda t\left[e^{\lambda t} + \lambda t\left(1 + \frac{\lambda t}{1!} + \frac{(\lambda t)^2}{2!} + \cdots\right)\right] \\
&= e^{-\lambda t}\,\lambda t\left[e^{\lambda t} + \lambda t\, e^{\lambda t}\right] \\
&= e^{-\lambda t}\, e^{\lambda t}\,\lambda t\,(1+\lambda t) \\
&= \lambda t\,(1+\lambda t) \\
\Rightarrow \quad E\big(X^2(t)\big) &= \lambda t + \lambda^2 t^2.
\end{aligned}$$

Therefore,

$$\begin{aligned}
\mathrm{Var}\,\big(X(t)\big) &= E\big(X^2(t)\big) - \big[E\big(X(t)\big)\big]^2 \\
&= \big(\lambda t + \lambda^2 t^2\big) - (\lambda t)^2 \\
&= \lambda t + \lambda^2 t^2 - \lambda^2 t^2 \\
&= \lambda t.
\end{aligned}$$

$$\mathrm{Var}\,\big(X(t)\big) = \lambda t, \qquad E\big(X^2(t)\big) = \lambda t + \lambda^2 t^2.$$

i.e., the variance of the Poisson process is also λt.

Note. Both the mean and the variance of the Poisson process are functions of time t; i.e., the number of occurrences of a Poisson process purely depends on the time interval of happening.

3 Problems on the Number of Occurrences

Problem 1

Messages arrive at a telegraph office in accordance with the laws of a Poisson process with a mean rate of 3 messages per hour. What is the probability that no message will arrive during the time interval 8 AM to 12 noon?

Solution. Given: the number of messages arriving at a telegraph office follows a Poisson process (PP) with a mean rate of 3 messages/hour.

We know that the mean of the PP is λt, where λ is the average rate of occurrences in unit time.

$$\Rightarrow \quad \boxed{\lambda = 3 \text{ per hour}}$$

$\therefore$ The pdf of the number of messages arrived in $(0, t)$ is given by

$$P\big(X(t) = n\big) = \frac{e^{-\lambda t}(\lambda t)^n}{n!} = \frac{e^{-3t}(3t)^n}{n!}, \qquad n = 0, 1, 2, \ldots$$

The time interval 8 AM to 12 noon is $t = 4$ hours.

$$\begin{aligned}
\therefore \; P(\text{no message arrives during 8 AM to 12 noon}) &= P\big(X(4) = 0\big) \\
&= \frac{e^{-3(4)}\,(3 \cdot 4)^0}{0!} \\
&= \frac{e^{-12} \cdot 1}{1} \\
&= e^{-12} \\
&= \frac{1}{e^{12}} \\
&= 6.144 \times 10^{-6}.
\end{aligned}$$

Problem 2

Suppose that the customers arrive at a bank according to a Poisson process with a mean rate of 3 per minute. Find the probability that during a time interval of 2 minutes (i) exactly 4 customers arrive, (ii) more than 4 customers arrive.

Solution. Given: the number of customers arriving at a bank follows a PP with $\lambda = 3$ per minute.

$$\therefore \text{ pdf is } \quad P\big(X(t) = n\big) = \frac{e^{-\lambda t}(\lambda t)^n}{n!} = \frac{e^{-3t}(3t)^n}{n!}, \qquad n = 0, 1, 2, \ldots$$

During the time interval of 2 minutes, i.e., $t = 2$, we have $\lambda t = 3(2) = 6$.

(i)

$$\begin{aligned}
P(\text{exactly 4 customers arrive}) &= P\big(X(t) = n\big)\Big|_{t=2,\ n=4} \\
&= P\big(X(2) = 4\big) \\
&= \frac{e^{-3(2)}\,(3 \cdot 2)^4}{4!} \\
&= \frac{e^{-6}\,6^4}{4!}
\end{aligned}$$

$$= \frac{e^{-6} \times 1296}{24}$$
$$= 54\, e^{-6}$$
$$= 0.1339.$$

(ii)

$$\begin{aligned} P(\text{more than 4 customers arrive}) &= P\big(X(2) > 4\big) \\ &= 1 - P\big(X(2) \le 4\big) \\ &= 1 - \Big[P\big(X(2)=0\big) + P\big(X(2)=1\big) + P\big(X(2)=2\big) \\ &\qquad + P\big(X(2)=3\big) + P\big(X(2)=4\big)\Big] \\ &= 1 - \left[\frac{e^{-6}6^0}{0!} + \frac{e^{-6}6^1}{1!} + \frac{e^{-6}6^2}{2!} + \frac{e^{-6}6^3}{3!} + \frac{e^{-6}6^4}{4!}\right] \\ &= 1 - e^{-6}\left[1 + 6 + \frac{36}{2} + \frac{216}{6} + \frac{1296}{24}\right] \\ &= 1 - e^{-6}\left[1 + 6 + 18 + 36 + 54\right] \\ &= 1 - 115\, e^{-6} \\ &= 1 - 0.2851 \\ \Rightarrow \quad P\big(X(2) > 4\big) &= 0.7149. \end{aligned}$$

Problem 3

Assume that the number of messages input to a communication channel in an interval of duration t seconds is a Poisson process with mean rate $\lambda = 0.3$. Compute the probability that (i) exactly 3 messages will arrive in a 10-second interval; (ii) the number of message arrivals in an interval of 5 seconds is between 3 and 7.

Solution. Given: the number of messages follows a PP with $\lambda = 0.3$ per second.

$$\therefore\ P\big(X(t)=n\big) = \frac{e^{-\lambda t}(\lambda t)^n}{n!} = \frac{e^{-0.3t}(0.3t)^n}{n!}, \qquad n = 0, 1, 2, \ldots$$

(i) Here $t = 10$ seconds, so $\lambda t = 0.3(10) = 3$.

$$\begin{aligned} P(\text{exactly 3 messages in 10 sec}) &= P\big(X(10)=3\big) \\ &= \frac{e^{-0.3(10)}\,(0.3\times 10)^3}{3!} \\ &= \frac{e^{-3}\,3^3}{3!} \\ &= \frac{27\, e^{-3}}{6} \\ &= 4.5\, e^{-3} \\ &= 4.5 \times 0.0498 \\ &= 0.2240. \end{aligned}$$

(ii) Here $t = 5$ seconds, so $\lambda t = 0.3(5) = 1.5$.

$$P(3 \le X(5) \le 7) = P\big(X(5)=3\big) + P\big(X(5)=4\big) + P\big(X(5)=5\big)$$

$$
\begin{aligned}
&\quad + P\big(X(5)=6\big) + P\big(X(5)=7\big) \\
&= \frac{e^{-1.5}(1.5)^3}{3!} + \frac{e^{-1.5}(1.5)^4}{4!} + \frac{e^{-1.5}(1.5)^5}{5!} + \frac{e^{-1.5}(1.5)^6}{6!} + \frac{e^{-1.5}(1.5)^7}{7!} \\
&= e^{-1.5}\left[\frac{3.375}{6} + \frac{5.0625}{24} + \frac{7.59375}{120} + \frac{11.390625}{720} + \frac{17.0859375}{5040}\right] \\
&= e^{-1.5}\left[0.5625 + 0.2109 + 0.0633 + 0.0158 + 0.0034\right] \\
&= 0.2231 \times 0.8559 \\
&= 0.1910.
\end{aligned}
$$

Problem 4 (Example)

A fisherman catches fish at a Poisson rate of 2 per hour. He starts fishing at 10 AM. Find the probability that he catches one fish by 10:30 AM and 3 fishes by noon.

Solution. Given $\lambda = 2$ per hour. Measure time from 10 AM. Then 10:30 AM corresponds to $t = \frac{1}{2}$ hour and noon corresponds to $t = 2$ hours.

"One fish by 10:30 AM and 3 fishes by noon" means one fish in $\big(0, \frac{1}{2}\big)$ and the remaining $3 - 1 = 2$ fishes in $\big(\frac{1}{2}, 2\big)$, an interval of length $2 - \frac{1}{2} = \frac{3}{2}$ hours. These intervals do not overlap, so (by the independent-increment property of the Poisson process)

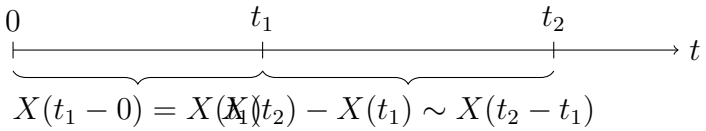
$$
\begin{aligned}
P\Big(X\big(\tfrac{1}{2}\big)=1,\ X(2)=3\Big) &= P\Big(X\big(\tfrac{1}{2}\big)=1\Big)\cdot P\Big(X(2)-X\big(\tfrac{1}{2}\big)=2\Big) \\
&= P\Big(X\big(\tfrac{1}{2}\big)=1\Big)\cdot P\Big(X\big(\tfrac{3}{2}\big)=2\Big) \\
&= \frac{e^{-2(1/2)}\left(2\cdot\frac{1}{2}\right)^1}{1!}\cdot\frac{e^{-2(3/2)}\left(2\cdot\frac{3}{2}\right)^2}{2!} \\
&= \frac{e^{-1}\cdot 1}{1}\cdot\frac{e^{-3}\cdot 9}{2} \\
&= 4.5\,e^{-4} \\
&= 4.5\times 0.0183 \\
&= 0.0824.
\end{aligned}
$$

4 Autocorrelation and Autocovariance of the Poisson Process

4.1 Result 3: Autocorrelation function

$$R(t_1,t_2) = \lambda^2 t_1 t_2 + \lambda \min\{t_1,t_2\}.$$

Proof. Let $t_2 \geq t_1$.



We know that the ACF is

$$R_X(t_1,t_2) = E\big(X(t_1)\,X(t_2)\big)$$

$$
\begin{aligned}
&= E\Big(X(t_1)\,\big[X(t_2) - X(t_1) + X(t_1)\big]\Big) \\
&= E\Big(X(t_1)\big[X(t_2) - X(t_1)\big] + X(t_1)\,X(t_1)\Big) \\
&= E\Big(X(t_1)\big[X(t_2) - X(t_1)\big]\Big) + E\big(X^2(t_1)\big) \\
&= E\Big(X(t_1 - 0)\cdot X(t_2 - t_1)\Big) + E\big(X^2(t_1)\big) \\
&\quad \text{(the counts in } (0, t_1) \text{ and } (t_1, t_2) \text{ depend only on the lengths } t_1 \text{ and } t_2 - t_1) \\
&= E\big(X(t_1 - 0)\big)\cdot E\big(X(t_2 - t_1)\big) + E\big(X^2(t_1)\big) \\
&\quad (\because (0, t_1) \text{ and } (t_1, t_2) \text{ are non-overlapping, so the counts are independent} \\
&\quad \text{by the postulates of the Poisson process}) \\
&= E\big(X(t_1)\big)\cdot E\big(X(t_2 - t_1)\big) + \lambda t_1 + \lambda^2 t_1^2 \\
&\quad (\because E(X^2(t)) = \lambda t + \lambda^2 t^2, \text{ by Result 2}) \\
&= \lambda t_1 \cdot \lambda(t_2 - t_1) + \lambda t_1 + \lambda^2 t_1^2 \\
&\quad (\because E(X(t)) = \lambda t, \text{ by Result 1}) \\
&= \lambda^2\big(t_1 t_2 - t_1^2\big) + \lambda t_1 + \lambda^2 t_1^2 \\
&= \lambda^2 t_1 t_2 - \lambda^2 t_1^2 + \lambda t_1 + \lambda^2 t_1^2 \\
\Rightarrow \quad R(t_1, t_2) &= \lambda^2 t_1 t_2 + \lambda t_1, \qquad t_2 \geq t_1.
\end{aligned}
$$

Similarly, if $t_1 \geq t_2$, then $R(t_1, t_2) = \lambda^2 t_1 t_2 + \lambda t_2$. Combining both cases,

$$\boxed{R(t_1, t_2) = \lambda^2 t_1 t_2 + \lambda \min\{t_1, t_2\}}$$

4.2 Result 4: Autocovariance function

$$C(t_1, t_2) = \lambda \min\{t_1, t_2\}.$$

Proof.

$$
\begin{aligned}
C(t_1, t_2) &= R(t_1, t_2) - E\big(X(t_1)\big)\,E\big(X(t_2)\big) \\
&= \lambda^2 t_1 t_2 + \lambda \min\{t_1, t_2\} - (\lambda t_1)(\lambda t_2) \\
&= \lambda^2 t_1 t_2 + \lambda \min\{t_1, t_2\} - \lambda^2 t_1 t_2 \\
&= \lambda \min\{t_1, t_2\}.
\end{aligned}
$$

5 Sum and Difference of Two Independent Poisson Processes

5.1 Result 5: Sum of two independent Poisson processes is again a Poisson process

Proof. Suppose $X(t)$ and $Y(t)$ are two independent random processes following Poisson processes with parameters λ_1 and λ_2 respectively. Then

$$E\big(X(t)\big) = \lambda_1 t \quad \text{and} \quad E\big(X^2(t)\big) = \lambda_1 t + \lambda_1^2 t^2.$$

Similarly,

$$E\big(Y(t)\big) = \lambda_2 t \quad \text{and} \quad E\big(Y^2(t)\big) = \lambda_2 t + \lambda_2^2 t^2.$$

Let $Z(t) = X(t) + Y(t)$. Then

$$\begin{aligned} E\big(Z(t)\big) &= E\big(X(t) + Y(t)\big) \\ &= E\big(X(t)\big) + E\big(Y(t)\big) \\ &= \lambda_1 t + \lambda_2 t \\ &= (\lambda_1 + \lambda_2)\, t \\ \Rightarrow \quad E\big(Z(t)\big) &= \lambda' t, \qquad \text{where } \lambda' = \lambda_1 + \lambda_2. \end{aligned}$$

Similarly,

$$\begin{aligned} E\big(Z^2(t)\big) &= E\big(X(t) + Y(t)\big)^2 \\ &= E\Big(X^2(t) + Y^2(t) + 2X(t)\, Y(t)\Big) \\ &= E\big(X^2(t)\big) + E\big(Y^2(t)\big) + 2\, E\big(X(t)\, Y(t)\big) \\ &= \big(\lambda_1 t + \lambda_1^2 t^2\big) + \big(\lambda_2 t + \lambda_2^2 t^2\big) + 2\, E\big(X(t)\big)\, E\big(Y(t)\big) \\ &\qquad (\because X(t) \text{ and } Y(t) \text{ are independent}) \\ &= \lambda_1 t + \lambda_1^2 t^2 + \lambda_2 t + \lambda_2^2 t^2 + 2(\lambda_1 t)(\lambda_2 t) \\ &= (\lambda_1 t + \lambda_2 t) + \big(\lambda_1^2 t^2 + \lambda_2^2 t^2 + 2\lambda_1 \lambda_2 t^2\big) \\ &= (\lambda_1 + \lambda_2)\, t + \big(\lambda_1^2 + \lambda_2^2 + 2\lambda_1 \lambda_2\big)\, t^2 \\ &= \lambda' t + (\lambda_1 + \lambda_2)^2\, t^2, \qquad \text{where } \lambda' = \lambda_1 + \lambda_2 \\ \Rightarrow \quad E\big(Z^2(t)\big) &= \lambda' t + \lambda'^2 t^2. \end{aligned}$$

i.e., the average results of $Z(t)$ coincide with the Poisson process results (with parameter $\lambda' = \lambda_1 + \lambda_2$). Hence $Z(t)$ also follows a Poisson process.

> The sum of two independent Poisson processes with parameters λ_1 and λ_2 follows a Poisson process with parameter $\lambda_1 + \lambda_2$.

5.2 Result 6: Difference of two independent Poisson processes is not a Poisson process

Proof. Let $X(t)$ and $Y(t)$ be as above and let $Z(t) = X(t) - Y(t)$. Then

$$\begin{aligned} E\big(Z(t)\big) &= E\big(X(t) - Y(t)\big) \\ &= E\big(X(t)\big) - E\big(Y(t)\big) \\ &= \lambda_1 t - \lambda_2 t \\ &= (\lambda_1 - \lambda_2)\, t \\ &= \lambda' t, \qquad \text{where } \lambda' = \lambda_1 - \lambda_2. \end{aligned}$$

And

$$\begin{aligned} E\big(Z^2(t)\big) &= E\big(X(t) - Y(t)\big)^2 \\ &= E\Big(X^2(t) + Y^2(t) - 2X(t)\, Y(t)\Big) \\ &= E\big(X^2(t)\big) + E\big(Y^2(t)\big) - 2\, E\big(X(t)\big)\, E\big(Y(t)\big) \qquad (\because \text{independence}) \\ &= \lambda_1 t + \lambda_1^2 t^2 + \lambda_2 t + \lambda_2^2 t^2 - 2(\lambda_1 t)(\lambda_2 t) \end{aligned}$$

$$
\begin{aligned}
&= (\lambda_1 + \lambda_2)\, t + \left(\lambda_1^2 + \lambda_2^2 - 2\lambda_1\lambda_2\right) t^2 \\
&= (\lambda_1 + \lambda_2)\, t + (\lambda_1 - \lambda_2)^2\, t^2 \\
\Rightarrow \quad E\left(Z^2(t)\right) &= \lambda'' t + \lambda'^2 t^2, \qquad \text{where } \lambda'' = \lambda_1 + \lambda_2 \neq \lambda' = \lambda_1 - \lambda_2.
\end{aligned}
$$

From $E\left(Z^2(t)\right)$ we see that the results do not coincide with the Poisson process result $\lambda' t + \lambda'^2 t^2$.

$\therefore$ The difference of two independent Poisson processes is **not** a Poisson process.

6 Inter-arrival Time of a Poisson Process

6.1 Result 7: The time interval between two successive occurrences follows an exponential distribution

Property

The interval time T between two successive occurrences of a Poisson process follows the exponential distribution

$$f(t) = \lambda e^{-\lambda t}, \quad t \geq 0, \qquad \text{with mean } \frac{1}{\lambda}.$$

Proof. Let T be the time interval between two successive occurrences E_i and E_{i+1} of the Poisson process. Assume that E_i is the event that happened at time t_i. Then

$$
\begin{aligned}
P(T > t) &= P\left(\text{time interval between successive occurrences } E_i,\, E_{i+1} \text{ exceeds } t\right) \\
&= P\left(E_{i+1} \text{ happens after the time interval } (t_i,\, t_i + t)\right) \\
&= P\left(\text{in the time interval } (t_i,\, t_i + t) \text{ no event has happened}\right) \\
&= P\left(X(t) = 0\right) \\
&= \frac{e^{-\lambda t}(\lambda t)^0}{0!} \\
\Rightarrow \quad P(T > t) &= e^{-\lambda t}.
\end{aligned}
$$

We know that the cdf of a random variable X is $F(x) = P(X \leq x)$. Hence the cdf of T is

$$
\begin{aligned}
F(t) &= P(T \leq t) \\
&= 1 - P(T > t) \\
&= 1 - e^{-\lambda t}.
\end{aligned}
$$

We know that the pdf is $f(x) = \dfrac{d}{dx} F(x)$. $\therefore$ The pdf of T is

$$
\begin{aligned}
f(t) &= \frac{d}{dt} F(t) \\
&= \frac{d}{dt}\left(1 - e^{-\lambda t}\right) \\
&= 0 - e^{-\lambda t}(-\lambda) \\
&= \lambda e^{-\lambda t}, \qquad t \geq 0.
\end{aligned}
$$

$\therefore$ The time interval between two successive occurrences of a Poisson process follows the exponential distribution.

Mean. We know that $E(X) = \int x\, f(x)\, dx$. Using Bernoulli's formula $\int u\, v\, dt = u\, v_1 - u'\, v_2 + u''\, v_3 - \cdots$ with $u = t$, $v = e^{-\lambda t}$, $v_1 = \dfrac{e^{-\lambda t}}{-\lambda}$, $v_2 = \dfrac{e^{-\lambda t}}{\lambda^2}$:

$$\begin{aligned}
\text{Mean} = E(T) &= \int_0^\infty t\, f(t)\, dt \\
&= \int_0^\infty t\, \lambda e^{-\lambda t}\, dt \\
&= \lambda \int_0^\infty t\, e^{-\lambda t}\, dt \\
&= \lambda \left[t\, \frac{e^{-\lambda t}}{-\lambda} - (1)\, \frac{e^{-\lambda t}}{\lambda^2} + 0 \right]_0^\infty \\
&= \lambda \left[-\frac{1}{\lambda}(0 - 0) - \frac{1}{\lambda^2}(0 - 1) \right] \\
&= \lambda \cdot \frac{1}{\lambda^2} \\
&= \frac{1}{\lambda}.
\end{aligned}$$

$$\Rightarrow \quad \boxed{\text{Mean} = \frac{1}{\lambda}}$$

6.2 Problems on inter-arrival time

Problem 5

Customers arrive at a counter in accordance with the Poisson process with a mean rate of 2 per minute. Find the probability that the interval time between two consecutive arrivals is (i) more than one minute, (ii) between 1 minute and 3 minutes, (iii) 4 minutes or less.

Solution. Let T be the time interval between two consecutive arrivals of customers to the counter. The arrivals follow a Poisson process with $\lambda = 2$ per minute. Then the pdf of T is

$$\begin{aligned}
f(t) &= \lambda e^{-\lambda t}, \quad t \geq 0 \\
f(t) &= 2e^{-2t}, \quad t \geq 0.
\end{aligned}$$

(i)

$$\begin{aligned}
P(\text{time interval is more than one minute}) &= P(T > 1) \\
&= \int_1^\infty f(t)\, dt \\
&= \int_1^\infty 2e^{-2t}\, dt \\
&= 2 \left[\frac{e^{-2t}}{-2} \right]_1^\infty
\end{aligned}$$

$$
\begin{aligned}
&= -\left[e^{-2t}\right]_1^{\infty} \\
&= -\left(0 - e^{-2}\right) \\
&= e^{-2} = \frac{1}{e^2} \\
&= 0.1353.
\end{aligned}
$$

(ii)

$$
\begin{aligned}
P(\text{between 1 min and 3 min}) &= P(1 < T < 3) \\
&= \int_1^3 f(t)\,dt \\
&= \int_1^3 2e^{-2t}\,dt \\
&= 2\left[\frac{e^{-2t}}{-2}\right]_1^3 \\
&= -\left(e^{-6} - e^{-2}\right) \\
&= \frac{1}{e^2} - \frac{1}{e^6} \\
&= 0.1353 - 0.0025 \\
&= 0.1329.
\end{aligned}
$$

(iii)

$$
\begin{aligned}
P(\text{4 minutes or less}) &= P(T \leq 4) \\
&= \int_0^4 f(t)\,dt \\
&= \int_0^4 2e^{-2t}\,dt \\
&= 2\left[\frac{e^{-2t}}{-2}\right]_0^4 \\
&= -\left[e^{-2t}\right]_0^4 \\
&= -\left(e^{-8} - 1\right) \\
&= 1 - \frac{1}{e^8} \\
&= 1 - 0.0003 \\
&= 0.9997.
\end{aligned}
$$

Problem 6

Consider a system with a Poisson job-arrival stream at an average rate of 60 per hour. Find the probability that the time interval between two successive job arrivals is (i) longer than 4 minutes, (ii) shorter than 8 minutes, (iii) between 2 and 6 minutes.

Solution. Given

$$
\begin{aligned}
&\lambda = 60 \text{ per hour} \\
\Rightarrow \quad &\lambda = \frac{60}{60} = 1 \text{ per minute.}
\end{aligned}
$$

The time interval T between two successive Poisson job arrivals follows the pdf

$$\begin{aligned} & f(t) = \lambda e^{-\lambda t}, \quad t \geq 0 \\ \Rightarrow \quad & f(t) = e^{-t}, \quad t \geq 0. \end{aligned}$$

(i)

$$\begin{aligned} P(T > 4) &= \int_4^\infty f(t)\, dt = \int_4^\infty e^{-t}\, dt = \left[\frac{e^{-t}}{-1}\right]_4^\infty \\ &= -\left(0 - e^{-4}\right) = e^{-4} \\ &= 0.0183. \end{aligned}$$

(ii)

$$\begin{aligned} P(T < 8) &= \int_0^8 f(t)\, dt = \int_0^8 e^{-t}\, dt = \left[\frac{e^{-t}}{-1}\right]_0^8 \\ &= -\left(e^{-8} - 1\right) = 1 - e^{-8} \\ &= 1 - 0.0003 \\ &= 0.9997. \end{aligned}$$

(iii)

$$\begin{aligned} P(2 < T < 6) &= \int_2^6 f(t)\, dt = \int_2^6 e^{-t}\, dt = \left[\frac{e^{-t}}{-1}\right]_2^6 \\ &= -\left(e^{-6} - e^{-2}\right) = e^{-2} - e^{-6} \\ &= 0.1353 - 0.0025 \\ &= 0.1329. \end{aligned}$$

Problem 7

Suppose customers arrive at a counter independently from two different sources, where arrivals occur in accordance with the Poisson process with a mean rate of 6 per hour from the first source and 4 per hour from the second source. Find the mean interval time between two successive arrivals to the counter.

Solution. Here customers arrive at a counter from two independent sources, each following a Poisson process, such that

$$\begin{aligned} \lambda_1 &= 6 \text{ per hour (from the first source)}, \\ \lambda_2 &= 4 \text{ per hour (from the second source)}. \end{aligned}$$

$\therefore$ At that counter, customer arrival is the sum of two independent Poisson processes.

$\therefore$ It will also follow a Poisson process (by Result 5) with

$$\begin{aligned} \lambda &= \lambda_1 + \lambda_2 \\ &= 6 + 4 \\ &= 10 \text{ per hour}, \end{aligned}$$

i.e., customer arrival at the counter has $\lambda = 10$ per hour.

$\therefore$ The mean interval time between two successive arrivals to the counter is $\frac{1}{\lambda}$ (by Result 7), i.e.,

$$\boxed{\text{Mean interval time} = \frac{1}{10} \text{ hour} = 6 \text{ minutes}.}$$

7 Recorded Occurrences of a Poisson Process

Definition. Suppose the number of occurrences of an event E follows a Poisson process with parameter λ. If each occurrence of the event E has a constant probability p of being recorded, then the number of recorded occurrences $N(t)$ in the time interval of length t also follows a Poisson process with parameter λp.

Hence the pdf of the number of recorded occurrences is given by

$$P\big(N(t)=n\big)=\frac{e^{-\lambda pt}(\lambda pt)^n}{n!}, \qquad n=0,1,2,\ldots$$

Problem 8

A radioactive source emits particles at a rate of 5 per minute in accordance with a Poisson process. Each particle emitted has a constant probability 0.6 of being recorded. Find the probability that 10 particles are recorded in a 4-minute period.

Solution. Based on the given conditions, we have

$$\lambda = 5 \text{ per minute},$$
$$p = 0.6.$$

$\therefore$ The number of recorded particles in a time interval of t takes the pdf

$$P\big(N(t)=n\big)=\frac{e^{-\lambda pt}(\lambda pt)^n}{n!}, \qquad n=0,1,2,\ldots$$
$$\Rightarrow \quad P\big(N(t)=n\big)=\frac{e^{-3t}(3t)^n}{n!}, \qquad n=0,1,2,\ldots \qquad (\because\ \lambda p = 5\times 0.6 = 3)$$

$$\therefore\ P(\text{10 particles are recorded in a 4-minute period}) = P\big(N(4)=10\big) \qquad (\because t=4,\ n=10)$$
$$= \frac{e^{-3(4)}\,(3\cdot 4)^{10}}{10!}$$
$$= \frac{e^{-12}\,(12)^{10}}{10!}$$
$$= 0.1048.$$

8 Summary: Three Types of pdfs in the Poisson Process

In the Poisson process we have 3 types of pdfs

1. $P\big(X(t)=n\big)=\dfrac{e^{-\lambda t}(\lambda t)^n}{n!}, \quad n=0,1,2,\ldots$
   Used for the number of occurrences in the time interval of t.
2. $f(t)=\lambda e^{-\lambda t}, \quad t\geq 0$
   Used for finding the probability of the time interval between two successive occurrences of the Poisson process.
3. $P\big(N(t)=n\big)=\dfrac{e^{-\lambda pt}(\lambda pt)^n}{n!}, \quad n=0,1,2,\ldots$
   Used for the probability of the number of recorded occurrences in t.

9 Pure Birth Process

Let $X(t)$ denote the number of events (births) that have occurred in $(0,t)$ and let

$$P_n(t)=P\big(X(t)=n\big), \qquad n=0,1,2,\ldots$$

We generalise the Poisson postulates by allowing the birth rate to depend on the present state n: when $X(t)=n$, the rate of a birth is λ_n.

9.1 Postulates

Consider an infinitesimal interval $(t,t+h)$ of length h. Given that n events have occurred in $(0,t)$,

$$P\big(1 \text{ event in } (t,t+h)\,\big|\,X(t)=n\big)=\lambda_n h+o(h),$$
$$P\big(2 \text{ or more events in } (t,t+h)\,\big|\,X(t)=n\big)=o(h).$$

Write $p_k=P\big(k \text{ events in } (t,t+h)\,\big|\,X(t)=n\big)$. Since the probabilities of $0,1,2,\ldots$ events in $(t,t+h)$ add up to 1,

$$\sum_{k=0}^{\infty}p_k=1$$
$$p_0+p_1+\sum_{k=2}^{\infty}p_k=1$$
$$\begin{aligned}p_0&=1-p_1-\sum_{k=2}^{\infty}p_k\\&=1-\big(\lambda_n h+o(h)\big)-o(h)\\&=1-\lambda_n h+o(h).\end{aligned}$$

Hence

$$P\big(k \text{ events in } (t,t+h)\,\big|\,X(t)=n\big)=\begin{cases}1-\lambda_n h+o(h), & k=0,\\ \lambda_n h+o(h), & k=1,\\ o(h), & k\geq 2.\end{cases} \tag{A}$$

These probabilities depend on the present state n and on h, but *not* on t.

9.2 Difference–differential equations

Now consider totally n events in $(0, t+h)$, i.e., $P_n(t+h)$. Splitting $(0, t+h)$ as $(0,t)$ and $(t, t+h)$, this can happen in the following mutually exclusive ways:

(a) n events in $(0,t)$ and no event in $(t, t+h)$;

(b) $n-1$ events in $(0,t)$ and 1 event in $(t, t+h)$;

(c) $n-k$ events in $(0,t)$ and $k \geq 2$ events in $(t, t+h)$.

Using $P(A \cup B) = P(A) + P(B)$ for mutually exclusive events and $P(A \cap B) = P(A)\,P(B \mid A)$,

$$\begin{aligned} P_n(t+h) &= P_n(t)\,P\big(0 \text{ events} \mid X(t) = n\big) + P_{n-1}(t)\,P\big(1 \text{ event} \mid X(t) = n-1\big) \\ &\quad + \sum_{k \geq 2} P_{n-k}(t)\,P\big(k \text{ events} \mid X(t) = n-k\big) \\ &= P_n(t)\big(1 - \lambda_n h + o(h)\big) + P_{n-1}(t)\big(\lambda_{n-1} h + o(h)\big) + o(h) \qquad (\because \text{by (A)}) \\ &= P_n(t) - \lambda_n h\,P_n(t) + \lambda_{n-1} h\,P_{n-1}(t) + o(h) \qquad (\because P_n(t), P_{n-1}(t) \leq 1) \end{aligned}$$

Subtracting $P_n(t)$ from both sides,

$$P_n(t+h) - P_n(t) = -\lambda_n h\,P_n(t) + \lambda_{n-1} h\,P_{n-1}(t) + o(h).$$

Dividing by h,

$$\frac{P_n(t+h) - P_n(t)}{h} = -\lambda_n P_n(t) + \lambda_{n-1} P_{n-1}(t) + \frac{o(h)}{h}.$$

Taking the limit as $h \to 0$,

$$\begin{aligned} \lim_{h \to 0} \frac{P_n(t+h) - P_n(t)}{h} &= -\lambda_n P_n(t) + \lambda_{n-1} P_{n-1}(t) + \lim_{h \to 0} \frac{o(h)}{h} \\ P_n'(t) &= -\lambda_n P_n(t) + \lambda_{n-1} P_{n-1}(t) + 0 \qquad (\because \lim_{h \to 0} o(h)/h = 0) \end{aligned}$$

When $n = 0$, only way (a) is possible (we cannot have -1 events in $(0,t)$, i.e., $P_{-1}(t) = 0$):

$$\begin{aligned} P_0(t+h) &= P_0(t)\big(1 - \lambda_0 h + o(h)\big) \\ P_0(t+h) - P_0(t) &= -\lambda_0 h\,P_0(t) + o(h) \\ \frac{P_0(t+h) - P_0(t)}{h} &= -\lambda_0 P_0(t) + \frac{o(h)}{h} \\ P_0'(t) &= -\lambda_0 P_0(t). \end{aligned}$$

Pure Birth Process

$$\begin{aligned} P_n'(t) &= -\lambda_n P_n(t) + \lambda_{n-1} P_{n-1}(t), \qquad n \geq 1, \qquad \text{(PB)} \\ P_0'(t) &= -\lambda_0 P_0(t). \end{aligned}$$

For given initial conditions, explicit expressions for $P_n(t)$ can be obtained from (PB). The process is called the **pure birth process**. Two important special cases are:

- $\lambda_n = \lambda$ (constant) $\Rightarrow$ the **Poisson process**;
- $\lambda_n = n\lambda \Rightarrow$ the **Yule–Furry process**.

9.3 Result 8: Derivation of the Poisson law from the postulates ($\lambda_n = \lambda$)

If $\lambda_n = \lambda$ for all n and $X(0) = 0$, then $P_n(t) = \dfrac{e^{-\lambda t}(\lambda t)^n}{n!}$, $n = 0, 1, 2, \ldots$

Proof. Putting $\lambda_n = \lambda$ in (PB),

$$\begin{aligned} P_n'(t) &= -\lambda P_n(t) + \lambda P_{n-1}(t), \qquad n \geq 1, \\ P_0'(t) &= -\lambda P_0(t). \end{aligned}$$

Since $X(0) = 0$, the initial conditions are $P_0(0) = 1$ and $P_n(0) = 0$ for $n \geq 1$.

For $n = 0$:

$$\begin{aligned} \frac{dP_0}{dt} &= -\lambda P_0(t) \\ \frac{dP_0}{P_0} &= -\lambda\, dt \\ \log_e P_0(t) &= -\lambda t + c \qquad (\because \text{ on integration }) \\ P_0(t) &= Ce^{-\lambda t}, \qquad \text{where } C = e^c. \end{aligned}$$

Substituting $P_0(0) = 1$: $1 = Ce^0 \Rightarrow C = 1$.

$$\therefore\ P_0(t) = e^{-\lambda t}.$$

For $n = 1$:

$$\begin{aligned} P_1'(t) &= -\lambda P_1(t) + \lambda P_0(t) \\ P_1'(t) + \lambda P_1(t) &= \lambda e^{-\lambda t}. \end{aligned}$$

This is a linear equation $\dfrac{dy}{dt} + Py = Q$ with $y = P_1(t)$, $P = \lambda$, $Q = \lambda e^{-\lambda t}$, whose solution is $y\, e^{\int P\, dt} = \int Q\, e^{\int P\, dt}\, dt + C$. Here the integrating factor is $e^{\int \lambda\, dt} = e^{\lambda t}$.

$$\begin{aligned} P_1(t)\, e^{\lambda t} &= \int \lambda e^{-\lambda t}\, e^{\lambda t}\, dt + C \\ &= \int \lambda\, dt + C \\ &= \lambda t + C. \end{aligned}$$

Substituting $P_1(0) = 0$: $0 = 0 + C \Rightarrow C = 0$.

$$\therefore\ P_1(t) = \lambda t\, e^{-\lambda t} = \frac{e^{-\lambda t}(\lambda t)^1}{1!}.$$

For $n = 2$:

$$\begin{aligned} P_2'(t) + \lambda P_2(t) = \lambda P_1(t) &= \lambda^2 t\, e^{-\lambda t} \\ P_2(t)\, e^{\lambda t} &= \int \lambda^2 t\, e^{-\lambda t}\, e^{\lambda t}\, dt + C \\ &= \int \lambda^2 t\, dt + C \end{aligned}$$

$$= \frac{\lambda^2 t^2}{2} + C.$$

Substituting $P_2(0) = 0$ gives $C = 0$.

$$\therefore\ P_2(t) = \frac{e^{-\lambda t}(\lambda t)^2}{2!}.$$

General n (by induction): Assume $P_{n-1}(t) = \dfrac{e^{-\lambda t}(\lambda t)^{n-1}}{(n-1)!}$. Then

$$\begin{aligned}
P_n'(t) + \lambda P_n(t) = \lambda P_{n-1}(t) &= \frac{\lambda^n t^{n-1} e^{-\lambda t}}{(n-1)!} \\
P_n(t)\, e^{\lambda t} &= \int \frac{\lambda^n t^{n-1} e^{-\lambda t}}{(n-1)!}\, e^{\lambda t}\, dt + C \\
&= \frac{\lambda^n}{(n-1)!} \int t^{n-1}\, dt + C \\
&= \frac{\lambda^n}{(n-1)!} \cdot \frac{t^n}{n} + C \\
&= \frac{(\lambda t)^n}{n!} + C.
\end{aligned}$$

Substituting $P_n(0) = 0$ gives $C = 0$. Therefore

$$\boxed{P_n(t) = P\big(X(t) = n\big) = \frac{e^{-\lambda t}(\lambda t)^n}{n!}, \qquad n = 0, 1, 2, \ldots}$$

which is the probability law of the Poisson process used in Sections 1–8.

10 Yule–Furry Process

Consider a population whose members are physical or biological entities. Suppose that members can give birth to new members but cannot die. Each member, independently of the others, gives birth in $(t, t+h)$ with probability $\lambda h + o(h)$.

Let $X(t)$ denote the total number of members at time t and $P_n(t) = P\big(X(t) = n\big)$. If $X(t) = n$, then

$$\begin{aligned}
P(\text{one birth in } (t, t+h) \,|\, X(t) = n) &= n\big(\lambda h + o(h)\big)\big(1 - \lambda h + o(h)\big)^{n-1} \\
&\qquad \text{(one of the } n \text{ members gives birth,} \\
&\qquad\ \ \text{the other } n-1 \text{ do not)} \\
&= n\big(\lambda h + o(h)\big)\big(1 - (n-1)\lambda h + o(h)\big) \\
&= n\lambda h + o(h).
\end{aligned}$$

Hence the Yule–Furry process is a pure birth process with $\lambda_n = n\lambda$. Putting $\lambda_n = n\lambda$ in (PB), we get

$$\begin{aligned}
P_n'(t) &= -n\lambda P_n(t) + (n-1)\lambda P_{n-1}(t), \qquad n \geq 1, \\
P_0'(t) &= -0 \cdot \lambda P_0(t) = 0.
\end{aligned} \tag{YF}$$

Suppose the process starts with one member, i.e., the initial conditions are

$$P_1(0) = 1 \quad \text{and} \quad P_i(0) = 0,\ i \neq 1.$$

For $n = 0$: $P_0'(t) = 0 \Rightarrow P_0(t) = \text{constant} = P_0(0) = 0$.

For $n = 1$: From (YF),

$$\begin{aligned} P_1'(t) &= -\lambda P_1(t) + 0 \cdot \lambda P_0(t) \\ \frac{dP_1}{dt} &= -\lambda P_1(t) \\ \frac{dP_1}{P_1} &= -\lambda\, dt \\ \log_e P_1(t) &= -\lambda t + c \qquad (\because \text{on integration}) \\ P_1(t) &= Ce^{-\lambda t}, \qquad \text{where } C = e^c. \end{aligned}$$

Substituting $P_1(0) = 1$ gives $C = 1$.

$$\therefore\ P_1(t) = e^{-\lambda t}. \tag{YF1}$$

For $n = 2$: From (YF),

$$\begin{aligned} P_2'(t) &= -2\lambda P_2(t) + \lambda P_1(t) \\ P_2'(t) + 2\lambda P_2(t) &= \lambda e^{-\lambda t} \qquad (\because \text{by (YF1)}) \end{aligned}$$

The above equation is of the form $\dfrac{dy}{dt} + Py = Q$, where $y = P_2(t)$, $P = 2\lambda$, $Q = \lambda e^{-\lambda t}$. Its solution is

$$y\, e^{\int P\, dt} = \int Q\, e^{\int P\, dt}\, dt + C.$$

i.e.,

$$\begin{aligned} P_2(t)\, e^{\int 2\lambda\, dt} &= \int \lambda e^{-\lambda t}\, e^{\int 2\lambda\, dt}\, dt + C \\ P_2(t)\, e^{2\lambda t} &= \int \lambda e^{-\lambda t}\, e^{2\lambda t}\, dt + C \\ &= \int \lambda e^{\lambda t}\, dt + C \\ &= e^{\lambda t} + C. \end{aligned}$$

Substituting the initial condition $P_2(0) = 0$: $0 = 1 + C \Rightarrow C = -1$.

$$\begin{aligned} P_2(t)\, e^{2\lambda t} &= e^{\lambda t} - 1 \\ P_2(t) &= e^{-2\lambda t}\big(e^{\lambda t} - 1\big) \\ &= e^{-\lambda t}\, e^{-\lambda t}\big(e^{\lambda t} - 1\big) \\ P_2(t) &= e^{-\lambda t}\big(1 - e^{-\lambda t}\big). \end{aligned}$$

General n (by induction): Assume $P_{n-1}(t) = e^{-\lambda t}\big(1 - e^{-\lambda t}\big)^{n-2}$ for some $n \geq 2$. Then

$$P_n'(t) + n\lambda P_n(t) = (n-1)\lambda P_{n-1}(t) = (n-1)\lambda\, e^{-\lambda t}\big(1 - e^{-\lambda t}\big)^{n-2}.$$

The integrating factor is $e^{n\lambda t}$, so

$$P_n(t)\, e^{n\lambda t} = \int (n-1)\lambda\, e^{-\lambda t}\big(1 - e^{-\lambda t}\big)^{n-2} e^{n\lambda t}\, dt + C$$

$$
\begin{aligned}
&= \int (n-1)\lambda\, e^{\lambda t}\, e^{(n-2)\lambda t}\big(1-e^{-\lambda t}\big)^{n-2}\, dt + C \\
&= \int (n-1)\lambda\, e^{\lambda t}\big(e^{\lambda t}-1\big)^{n-2}\, dt + C \\
&= \big(e^{\lambda t}-1\big)^{n-1} + C \qquad (\because \tfrac{d}{dt}(e^{\lambda t}-1)^{n-1} = (n-1)(e^{\lambda t}-1)^{n-2}\lambda e^{\lambda t})
\end{aligned}
$$

Substituting $P_n(0) = 0$: $0 = (1-1)^{n-1} + C \Rightarrow C = 0$.

$$
\begin{aligned}
P_n(t) &= e^{-n\lambda t}\big(e^{\lambda t}-1\big)^{n-1} \\
&= e^{-\lambda t}\, e^{-(n-1)\lambda t}\big(e^{\lambda t}-1\big)^{n-1} \\
&= e^{-\lambda t}\big(1-e^{-\lambda t}\big)^{n-1}.
\end{aligned}
$$

Yule–Furry Process

$$P_n(t) = e^{-\lambda t}\big(1-e^{-\lambda t}\big)^{n-1}, \quad n \geq 1, \qquad \text{and} \qquad P_0(t) = 0.$$

i.e., $X(t)$ has a geometric distribution $P_n(t) = p\,q^{n-1}$, $n = 1, 2, \ldots$, with $p = e^{-\lambda t}$ and $q = 1 - p = 1 - e^{-\lambda t}$.

Check:

$$\sum_{n=1}^{\infty} P_n(t) = \sum_{n=1}^{\infty} p\,q^{n-1} = p\big(1 + q + q^2 + \cdots\big) = p\,(1-q)^{-1} = \frac{p}{p} = 1.$$

10.1 Mean and variance of the Yule–Furry process

Mean.

$$
\begin{aligned}
E\big(X(t)\big) &= \sum_{n=1}^{\infty} n\, P_n(t) \\
&= \sum_{n=1}^{\infty} n\, p\, q^{n-1} \\
&= p\big(1 + 2q + 3q^2 + 4q^3 + \cdots\big) \\
&= p\,(1-q)^{-2} \\
&= \frac{p}{p^2} = \frac{1}{p} \\
&= \frac{1}{e^{-\lambda t}} \\
&= e^{\lambda t}.
\end{aligned}
$$

Variance. Write $n^2 = n(n-1) + n$. Then

$$
\begin{aligned}
\sum_{n=1}^{\infty} n(n-1)\, p\, q^{n-1} &= p\, q \sum_{n=2}^{\infty} n(n-1)\, q^{n-2} \\
&= p\, q\big(2 \cdot 1 + 3 \cdot 2\, q + 4 \cdot 3\, q^2 + \cdots\big) \\
&= 2p\, q\big(1 + 3q + 6q^2 + \cdots\big) \\
&= 2p\, q\,(1-q)^{-3}
\end{aligned}
$$

$$= \frac{2p\,q}{p^3} = \frac{2q}{p^2}.$$

$$\begin{aligned} E\big(X^2(t)\big) &= \sum_{n=1}^{\infty} n^2\, p\, q^{\,n-1} \\ &= \sum_{n=1}^{\infty} n(n-1)\, p\, q^{\,n-1} + \sum_{n=1}^{\infty} n\, p\, q^{\,n-1} \\ &= \frac{2q}{p^2} + \frac{1}{p}. \end{aligned}$$

$$\begin{aligned} \text{Var}\,\big(X(t)\big) &= E\big(X^2(t)\big) - \big[E\big(X(t)\big)\big]^2 \\ &= \frac{2q}{p^2} + \frac{1}{p} - \frac{1}{p^2} \\ &= \frac{2q + p - 1}{p^2} \\ &= \frac{2q - q}{p^2} \qquad (\because\, p - 1 = -q\,) \\ &= \frac{q}{p^2} \\ &= \frac{1 - e^{-\lambda t}}{e^{-2\lambda t}} \\ &= e^{2\lambda t}\big(1 - e^{-\lambda t}\big) \\ &= e^{2\lambda t} - e^{\lambda t}. \end{aligned}$$

$$E\big(X(t)\big) = e^{\lambda t}, \qquad \text{Var}\,\big(X(t)\big) = e^{2\lambda t} - e^{\lambda t} = e^{\lambda t}\big(e^{\lambda t} - 1\big).$$

11 Birth and Death Process

In a birth and death process, suppose λ represents the mean rate of births and μ represents the mean rate of deaths, and $P_n(t)$ represents the probability that there are n individuals in the population at time t, i.e., $P_n(t) = P\big(X(t) = n\big)$.

Based on the birth and death parameters λ and μ, we have the following (as in the Poisson postulates), births and deaths in $(t, t+h)$ being independent:

1. $P(1 \text{ birth in } (t, t+h)) = \lambda h + o(h)$
2. $P(2 \text{ or more births in } (t, t+h)) = o(h)$
3. $P(0 \text{ births in } (t, t+h)) = 1 - \lambda h + o(h)$
4. $P(1 \text{ death in } (t, t+h)) = \mu h + o(h) \quad (\text{when } n \geq 1)$
5. $P(2 \text{ or more deaths in } (t, t+h)) = o(h)$
6. $P(0 \text{ deaths in } (t, t+h)) = 1 - \mu h + o(h)$

When the population size is 0, no death can occur.

11.1 Equation for $n \geq 1$

The probability that there are n individuals at time $t+h$ is $P_n(t+h)$. Splitting $(0, t+h)$ as $(0,t)$ and $(t, t+h)$, this can be attained in the following mutually exclusive ways $A_1, \ldots, A_5$:

A_1: n individuals at time t and no births and no deaths in $(t, t+h)$:

$$\begin{aligned} P(A_1) &= P_n(t)\big(1-\lambda h+o(h)\big)\big(1-\mu h+o(h)\big) \\ &= P_n(t)\big(1-\lambda h-\mu h+\lambda\mu h^2+o(h)\big) \\ &= P_n(t)\big(1-(\lambda+\mu)h\big)+o(h) \qquad (\because \lambda\mu h^2 = o(h)\,) \end{aligned}$$

A_2: n individuals at time t and 1 birth and 1 death in $(t, t+h)$:

$$\begin{aligned} P(A_2) &= P_n(t)\big(\lambda h+o(h)\big)\big(\mu h+o(h)\big) \\ &= P_n(t)\big(\lambda\mu h^2+o(h)\big) \\ &= o(h) \end{aligned}$$

A_3: $n-1$ individuals at time t and 1 birth and 0 deaths in $(t, t+h)$:

$$\begin{aligned} P(A_3) &= P_{n-1}(t)\big(\lambda h+o(h)\big)\big(1-\mu h+o(h)\big) \\ &= P_{n-1}(t)\big(\lambda h-\lambda\mu h^2+o(h)\big) \\ &= \lambda h\, P_{n-1}(t)+o(h) \end{aligned}$$

A_4: $n+1$ individuals at time t and 0 births and 1 death in $(t, t+h)$:

$$\begin{aligned} P(A_4) &= P_{n+1}(t)\big(1-\lambda h+o(h)\big)\big(\mu h+o(h)\big) \\ &= P_{n+1}(t)\big(\mu h-\lambda\mu h^2+o(h)\big) \\ &= \mu h\, P_{n+1}(t)+o(h) \end{aligned}$$

A_5: any other combination, which needs 2 or more events (births and/or deaths) in $(t, t+h)$:

$$P(A_5) = o(h).$$

Thus

$$\{X(t+h)=n\} = A_1 \cup A_2 \cup A_3 \cup A_4 \cup A_5,$$

and since the A_i are mutually exclusive,

$$\begin{aligned} P_n(t+h) &= P(A_1)+P(A_2)+P(A_3)+P(A_4)+P(A_5) \\ &= \big[P_n(t)\big(1-(\lambda+\mu)h\big)+o(h)\big]+o(h)+\big[\lambda h\, P_{n-1}(t)+o(h)\big] \\ &\quad +\big[\mu h\, P_{n+1}(t)+o(h)\big]+o(h) \\ &= P_n(t)-(\lambda+\mu)h\, P_n(t)+\lambda h\, P_{n-1}(t)+\mu h\, P_{n+1}(t)+o(h). \end{aligned}$$

Subtracting $P_n(t)$ from both sides,

$$P_n(t+h)-P_n(t) = \lambda h\, P_{n-1}(t)-(\lambda+\mu)h\, P_n(t)+\mu h\, P_{n+1}(t)+o(h).$$

Dividing by h,

$$\frac{P_n(t+h)-P_n(t)}{h} = \lambda P_{n-1}(t)-(\lambda+\mu)P_n(t)+\mu P_{n+1}(t)+\frac{o(h)}{h}.$$

Taking the limit as $h \to 0$,

$$\begin{aligned} \lim_{h\to 0}\frac{P_n(t+h)-P_n(t)}{h} &= \lambda P_{n-1}(t)-(\lambda+\mu)P_n(t)+\mu P_{n+1}(t)+\lim_{h\to 0}\frac{o(h)}{h} \\ P_n'(t) &= \lambda P_{n-1}(t)-(\lambda+\mu)P_n(t)+\mu P_{n+1}(t), \qquad n \geq 1. \end{aligned} \tag{BD1}$$

11.2 Equation for $n = 0$

When $n = 0$, a death cannot occur (there is nobody to die) and $P_{-1}(t) = 0$. So $X(t+h) = 0$ only in the ways:

B_1: 0 individuals at time t and no birth in $(t, t+h)$:

$$P(B_1) = P_0(t)\big(1 - \lambda h + o(h)\big) = P_0(t) - \lambda h\, P_0(t) + o(h).$$

B_2: 1 individual at time t, 0 births and 1 death in $(t, t+h)$:

$$P(B_2) = P_1(t)\big(1 - \lambda h + o(h)\big)\big(\mu h + o(h)\big) = \mu h\, P_1(t) + o(h).$$

B_3: any other combination: $P(B_3) = o(h)$.

$$\begin{aligned}
P_0(t+h) &= P(B_1) + P(B_2) + P(B_3) \\
&= P_0(t) - \lambda h\, P_0(t) + \mu h\, P_1(t) + o(h) \\
P_0(t+h) - P_0(t) &= -\lambda h\, P_0(t) + \mu h\, P_1(t) + o(h) \\
\frac{P_0(t+h) - P_0(t)}{h} &= -\lambda P_0(t) + \mu P_1(t) + \frac{o(h)}{h} \\
P_0'(t) &= -\lambda P_0(t) + \mu P_1(t). \qquad \text{(BD2)}
\end{aligned}$$

Birth and Death Process

$$\begin{aligned}
P_n'(t) &= \lambda P_{n-1}(t) - (\lambda + \mu) P_n(t) + \mu P_{n+1}(t), \qquad n \geq 1, \\
P_0'(t) &= -\lambda P_0(t) + \mu P_1(t).
\end{aligned}$$

Equations (BD1) and (BD2) are called the **difference–differential equations** of the birth and death process. They give the rate of change of the probability $P_n(t)$ of having n individuals at time t.

Note.

1. In equation (BD2) the coefficient of $P_0(t)$ is $-\lambda$ and *not* $-(\lambda+\mu)$, since no death can take place when the population is empty.
2. If the rates depend on the state (λ_n, μ_n), the same argument gives

$$\begin{aligned}
P_n'(t) &= \lambda_{n-1} P_{n-1}(t) - (\lambda_n + \mu_n) P_n(t) + \mu_{n+1} P_{n+1}(t), \qquad n \geq 1, \\
P_0'(t) &= -\lambda_0 P_0(t) + \mu_1 P_1(t).
\end{aligned}$$

   Putting $\mu_n = 0$ gives the pure birth process (PB).

12 Renewal Process

In the Poisson process, the intervals between two successive events are independent and identically distributed (i.i.d.) with the (negative) exponential distribution (Result 7). If the intervals are i.i.d. but not necessarily exponential, we get a generalisation of the Poisson process called the **renewal process**. The renewal process may be considered in continuous time or in discrete time.

Definition. Let X_i be the time between the $(i-1)$-th and the i-th event (renewal), $i = 1, 2, 3, \ldots$. If $\{X_i;\ i = 1, 2, 3, \ldots\}$ is a sequence of independent, non-negative, identically distributed random variables, then the process of events generated by $\{X_i\}$ is called a **renewal process** (or recurrent process). Let

$$S_0 = 0, \qquad S_n = X_1 + X_2 + \cdots + X_n \quad \text{(the time of the } n\text{-th renewal)},$$

and let $N(t)$ be the number of renewals in $(0, t]$. Then $\{N(t) : t \geq 0\}$ is the **renewal counting process**.

When each X_i is exponential with mean $1/\lambda$, $N(t)$ is the Poisson process.

12.1 Discrete state, discrete time renewal process

Consider a sequence of repeated trials with possible outcomes $E_j;\ j = 1, 2, 3, \ldots$. The trials need not be independent. We assume that the trials can be repeated indefinitely.

Suppose that we are interested in a certain outcome in a trial, or a pattern of outcomes in a number of trials. We denote this event by E^*. Whenever E^* occurs we say that a *renewal* has occurred. If it occurs at the n-th trial, we say that a renewal occurs at trial number n. Once a renewal has occurred, the trials are counted afresh from that trial onwards.

Renewal period

The interval between the occurrences of two successive renewals is called a **renewal period** of the process. Define

$$f_n = P\big(E^* \text{ occurs for the first time at the } n\text{-th trial}\big),$$
$$u_n = P\big(E^* \text{ occurs at the } n\text{-th trial, not necessarily for the first time}\big).$$

Then

$$f^* = \sum_{n=1}^{\infty} f_n$$

is the probability that a renewal occurs at some trial in a long sequence of trials. We have $f^* \leq 1$.

If $f^* = 1$, then $\{f_n\}$ is a probability distribution representing the distribution of the length T of the renewal period, i.e., $P(T = n) = f_n$, and its mean $\mu = \sum_{n=1}^{\infty} n\, f_n$ is the **mean renewal period**.

The renewal event E^* is said to be **persistent** (recurrent) if $f^* = 1$ and **transient** if $f^* < 1$.

Example

Consider a fair die throwing experiment. Let E^* be the event that the face 4 appears. Show that E^* is persistent and find the mean renewal period.

Solution. In each throw,

$$P(E^* \text{ occurs in a trial}) = P(\text{getting a } 4) = \frac{1}{6},$$
$$P(E^* \text{ does not occur in a trial}) = P(\text{not getting a } 4) = \frac{5}{6}.$$

E^* occurs for the first time at the n-th trial if the first $n-1$ throws are not 4 and the n-th throw is 4:

$$f_n = \left(\frac{5}{6}\right)^{n-1}\left(\frac{1}{6}\right), \qquad n = 1, 2, 3, \ldots$$

$$\begin{aligned}
f^* &= \sum_{n=1}^{\infty} f_n = f_1 + f_2 + f_3 + \cdots \\
&= \left(\frac{1}{6}\right) + \left(\frac{5}{6}\right)\left(\frac{1}{6}\right) + \left(\frac{5}{6}\right)^2\left(\frac{1}{6}\right) + \cdots \\
&= \left(\frac{1}{6}\right)\left[1 + \left(\frac{5}{6}\right) + \left(\frac{5}{6}\right)^2 + \cdots\right] \\
&= \left(\frac{1}{6}\right)\left[1 - \frac{5}{6}\right]^{-1} \\
&= \left(\frac{1}{6}\right)\left(\frac{1}{6}\right)^{-1} \\
&= \frac{1}{6} \times 6 \\
f^* &= 1.
\end{aligned}$$

Thus the renewal event is persistent and the renewal period T has a geometric distribution. Its mean is

$$\begin{aligned}
\mu = E(T) &= \sum_{n=1}^{\infty} n\, f_n \\
&= \left(\frac{1}{6}\right)\left[1 + 2\left(\frac{5}{6}\right) + 3\left(\frac{5}{6}\right)^2 + \cdots\right] \\
&= \left(\frac{1}{6}\right)\left[1 - \frac{5}{6}\right]^{-2} \\
&= \left(\frac{1}{6}\right)(36) \\
&= 6 \text{ throws.}
\end{aligned}$$

12.2 Discrete state, continuous parameter renewal process

Let $N(t)$ denote the number of renewals in the interval $(0, t]$. Then $\{N(t) : t \geq 0\}$ is a discrete state, continuous parameter random process called the **renewal counting process**.

Let $\{X_n;\ n = 1, 2, 3, \ldots\}$ be a sequence of non-negative, independent, identically distributed random variables with $P(X_n = 0) < 1$. Assume that they are continuous with distribution function $F(x)$ and density function $f(x) = F'(x)$, $x \geq 0$.

Let $S_0 = 0$, $S_n = X_1 + X_2 + \cdots + X_n$. Then S_k denotes the time from the beginning until the occurrence of the k-th renewal. Let $F_k(t) = P(S_k \leq t)$ denote the distribution function of S_k; in particular $F_1(t) = F(t)$. For notational convenience we define

$$F_0(t) = \begin{cases} 1, & t \geq 0, \\ 0, & t < 0. \end{cases}$$

Distribution of $N(t)$

We can relate the renewal random variable $N(t)$ to the random variables S_k as follows:

$$N(t) = n \iff S_n \le t < S_{n+1}.$$

Since $S_{n+1} \ge S_n$, the event $\{S_n \le t\}$ splits into the mutually exclusive events $\{S_n \le t < S_{n+1}\}$ and $\{S_{n+1} \le t\}$:

$$\begin{aligned} P(S_n \le t) &= P(S_n \le t < S_{n+1}) + P(S_{n+1} \le t) \\ \Rightarrow\ P(S_n \le t < S_{n+1}) &= P(S_n \le t) - P(S_{n+1} \le t). \end{aligned}$$

Therefore

$$\begin{aligned} P\big(N(t) = n\big) &= P(S_n \le t < S_{n+1}) \\ &= P(S_n \le t) - P(S_{n+1} \le t) \\ &= F_n(t) - F_{n+1}(t), \qquad n = 0, 1, 2, \ldots \end{aligned}$$

Renewal function $M(t)$

The **renewal function** $M(t)$ is the average number of renewals in the interval $(0, t]$, i.e., $M(t) = E\big(N(t)\big)$.

$$\begin{aligned} M(t) = E\big(N(t)\big) &= \sum_{n=0}^{\infty} n\, P\big(N(t) = n\big) \\ &= \sum_{n=0}^{\infty} n \left[F_n(t) - F_{n+1}(t)\right] \\ &= \sum_{n=0}^{\infty} n\, F_n(t) - \sum_{n=0}^{\infty} n\, F_{n+1}(t) \\ &= \big[0F_0(t) + 1F_1(t) + 2F_2(t) + 3F_3(t) + \cdots\big] \\ &\quad - \big[0F_1(t) + 1F_2(t) + 2F_3(t) + \cdots\big] \\ &= F_1(t) + (2-1)F_2(t) + (3-2)F_3(t) + \cdots \\ &= F_1(t) + F_2(t) + F_3(t) + \cdots \\ &= \sum_{n=1}^{\infty} F_n(t). \end{aligned}$$

$$M(t) = \sum_{n=1}^{\infty} F_n(t) = F(t) + \sum_{n=2}^{\infty} F_n(t).$$

Fundamental renewal equation

Since $S_{n+1} = S_n + X_{n+1}$ with S_n and X_{n+1} independent, conditioning on $X_{n+1} = x$,

$$\begin{aligned} F_{n+1}(t) &= P(S_n + X_{n+1} \le t) \\ &= \int_0^t P(S_n \le t - x)\, f(x)\, dx \end{aligned}$$

$$= \int_0^t F_n(t-x)\, f(x)\, dx,$$

i.e., F_{n+1} is the convolution of F_n with F. Hence

$$\begin{aligned}
M(t) &= F(t) + \sum_{n=2}^{\infty} F_n(t) \\
&= F(t) + \sum_{n=2}^{\infty} \int_0^t F_{n-1}(t-x)\, f(x)\, dx \\
&= F(t) + \int_0^t \left[\sum_{n=2}^{\infty} F_{n-1}(t-x)\right] f(x)\, dx \\
&= F(t) + \int_0^t \left[\sum_{k=1}^{\infty} F_k(t-x)\right] f(x)\, dx \qquad (\because \text{ put } k = n-1\,) \\
&= F(t) + \int_0^t M(t-x)\, f(x)\, dx.
\end{aligned}$$

Fundamental Renewal Equation

$$M(t) = F(t) + \int_0^t M(t-x)\, f(x)\, dx.$$

12.3 Solution of the renewal equation

Let $m(t) = M'(t)$ be the **renewal density**. Differentiating the fundamental renewal equation with respect to t (Leibniz rule), and using $M(0) = 0$,

$$\begin{aligned}
m(t) &= \frac{d}{dt}F(t) + \frac{d}{dt}\int_0^t M(t-x)\, f(x)\, dx \\
&= f(t) + M(t-t)\, f(t) + \int_0^t M'(t-x)\, f(x)\, dx \\
&= f(t) + M(0)\, f(t) + \int_0^t m(t-x)\, f(x)\, dx \\
&= f(t) + \int_0^t m(t-x)\, f(x)\, dx.
\end{aligned}$$

So we consider the renewal equation for the density:

$$m(t) = f(t) + \int_0^t m(t-x)\, f(x)\, dx.$$

Taking the Laplace transform on both sides, with $L_m(s) = L[m(t)]$ and $L_f(s) = L[f(t)]$,

$$\begin{aligned}
L[m(t)] &= L[f(t)] + L\left[\int_0^t m(t-x)\, f(x)\, dx\right] \\
L_m(s) &= L_f(s) + L\big[m * f\big] \\
L_m(s) &= L_f(s) + L_m(s)\, L_f(s) \qquad (\because \text{ convolution theorem}\,)
\end{aligned}$$

Solving for $L_m(s)$:

$$\begin{aligned} L_m(s) - L_m(s)\, L_f(s) &= L_f(s) \\ L_m(s)\big[1 - L_f(s)\big] &= L_f(s) \\ L_m(s) &= \frac{L_f(s)}{1 - L_f(s)}. \end{aligned}$$

Similarly, solving for $L_f(s)$:

$$\begin{aligned} L_m(s) &= L_f(s) + L_m(s)\, L_f(s) \\ L_m(s) &= L_f(s)\big[1 + L_m(s)\big] \\ L_f(s) &= \frac{L_m(s)}{1 + L_m(s)}. \end{aligned}$$

$$L_m(s) = \frac{L_f(s)}{1 - L_f(s)}, \qquad L_f(s) = \frac{L_m(s)}{1 + L_m(s)}.$$

Using the above two equations, $m(t)$ can be determined from $f(t)$ and conversely.

Example

If the inter-renewal times are exponential with $f(x) = \lambda e^{-\lambda x}$, $x \geq 0$, find the renewal density $m(t)$ and the renewal function $M(t)$.

Solution.

$$\begin{aligned} L_f(s) &= \int_0^\infty e^{-st}\, \lambda e^{-\lambda t}\, dt \\ &= \lambda \int_0^\infty e^{-(s+\lambda)t}\, dt \\ &= \lambda \left[\frac{e^{-(s+\lambda)t}}{-(s+\lambda)} \right]_0^\infty \\ &= \lambda \left[0 - \frac{1}{-(s+\lambda)} \right] \\ &= \frac{\lambda}{s+\lambda}. \end{aligned}$$

$$\begin{aligned} L_m(s) &= \frac{L_f(s)}{1 - L_f(s)} \\ &= \frac{\dfrac{\lambda}{s+\lambda}}{1 - \dfrac{\lambda}{s+\lambda}} \\ &= \frac{\dfrac{\lambda}{s+\lambda}}{\dfrac{s+\lambda-\lambda}{s+\lambda}} \\ &= \frac{\lambda}{s}. \end{aligned}$$

Taking the inverse Laplace transform, $m(t) = L^{-1}\left[\frac{\lambda}{s}\right] = \lambda$. Hence

$$M(t) = \int_0^t m(x)\,dx = \int_0^t \lambda\,dx = \lambda t,$$

which agrees with the mean of the Poisson process (Result 1), as it should, since exponential inter-renewal times give the Poisson process.

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